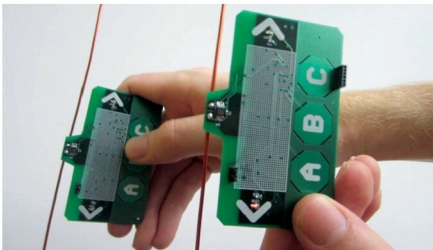


Turn your dumb TV into a smartTV

If you do not want to spend money on a new smartTV, there are ways to turn your present TV into one. For instance, WD TV Live Hub and iOmega ScreenPlay DX HD are two set-top box type media players for TV that are ideal for accessing web services like TuneIn, YouTube and Flickr. These have a nice software interface and a hard drive of 1TB size. WD TV Live Hub is available for ₹ 12,500 while iOmega ScreenPlay DX HD is available for ₹ 12,000.

Amketo offers EvoTV series of devices, costing ₹ 5500 to ₹ 10,500. This set-top box runs on Google's Android OS, which is an advantage because you can access Google's apps using Play Store. There is also an option to attach USB keyboard to make the experience more interactive. Another option, Apple TV, is a network media player that you can connect to TV for streaming pictures, music and videos from iPhone or iPad. Apple lets users subscribe to channels using iTunes.

Google's Chromecast is an easy-to-use HDMI device through which you can connect to Wi-Fi network, or link it to a laptop or mobile device to stream the content from these devices on TV. Chromecast also provides an option to stream YouTube videos on TV screen. Users can subscribe to services like BigFlix, NexGTV, Spuul and BoxTV to



Using ambient backscatter, these devices can interact with users and communicate with each other without using batteries. They exchange information by reflecting or absorbing pre-existing radio signals

stream their shows on TV. It is available on online stores for ₹ 3999. One can insert it in HDMI port of TV and even connect its microUSB port to TV's USB port for alternative source.

Technology to connect battery-free devices to Wi-Fi

A new technology will help connect battery-free devices to Wi-Fi and also allow the devices to store information about our daily activities, but everything will be done without any battery. The new communication system developed by researchers at the University of Washington uses radio frequency signals as a power source. Existing Wi-Fi infrastructure is being reused to provide Internet to all these devices. Embedded sensors will help in monitoring and tracking everything ranging from safety measures at bridges and health factors.

Providing Wi-Fi connectivity to the devices is not an easy job as conventional, low-power Wi-Fi re-

quires more power than usual. Hence an ultra-low-power tag prototype associated with an antenna and circuit has been developed by the researchers. This particular technology helps in communication with Wi-Fi enabled laptops or smartphones using a very negligible amount of power.

Wi-Fi signals moving between the router and a laptop or a smartphone help in activation of these tags. The data is encoded by these tags, and sometimes Wi-Fi routers' signals get reflected and sometimes they don't. As a result, the wireless signals get slightly changed. Laptops and smartphones, which are equipped with Wi-Fi, are able to detect these changes and data is received from the tag.

Efficient spray-on solar cells developed

Researchers from the University of Sheffield, led by Prof. David Lidzey, have found a new method of tapping solar energy. They used a crystalline organometal called Perovskite to make cheap and efficient solar cells. Perovskite, made of calcium titanate, is cheaper as compared to silicon. Silicon needs a layer of minimum 180 micro-



iOmega ScreenPlay DX HD